

Cover exactingness, definability, and conditional inconsistency

A cofinality barrier in the HCD hierarchy,
local covering obstructions, and an I0 comparison

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Main result. A γ -cover-exacting cardinal λ is regular in $\text{HCD}(\gamma^+)$, the inner model of sets hereditarily definable from γ^+ -complete ultrafilters on ordinals. This yields two different incompatibility results, depending on the position of a strongly compact cardinal.

Below the cover-exacting cardinal: a forced gap

If $\delta < \lambda$ is strongly compact and λ is γ -cover exacting, there is an unbounded $a \subseteq \lambda$ such that

$$a \in \text{HCD}(\delta), \quad \text{ot}(a) < \delta, \quad a \notin \text{HCD}(\gamma^+).$$

In fact, no member of $\text{HCD}(\gamma^+)$ of ambient size less than λ can contain a . Thus even a weak, explicitly specified local covering principle between these two canonical inner models is inconsistent with the configuration.

Above the cover bound: a proper ground is unavoidable

If $\kappa > \gamma$ is strongly compact, then $\text{HCD}(\kappa)$ is a proper ground of V in which λ is regular. Consequently,

$$\text{ZFC} + \text{GA} + \exists \lambda \leq \gamma < \kappa [\text{CEx}_\gamma(\lambda) \wedge \text{SC}(\kappa)]$$

is inconsistent. Here GA is the Ground Axiom. This is not a prohibition of cover exactingness alone or of its coexistence with a larger strongly compact cardinal.

Positive comparison. An ultraexacting cardinal can consistently be the first exacting cardinal, with every set of lower rank in HOD, while a single countable cofinal set witnesses strong failure of approximation and covering at the boundary. The full package is equiconsistent with I0.

Attribution and scope. The proofs below derive refinements from the Blue–Goldberg cover-exacting framework and Goldberg’s published HCD theorems. The affirmative package is a corollary of the existing Aguilera–Bagaria–Goldberg–Lücke forcing and equiconsistency results, not a new forcing construction. The bounded HCD obstruction is the principal proposed refinement; historical priority is not certified. The unrestricted strongly-compact-below-cover-exacting problem is not resolved here.

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Reading guide

Sections 3–5 prove the main definability barrier. Section 6 gives the local inconsistency theorem above a strongly compact cardinal. Section 7 gives the Ground Axiom incompatibility in the opposite ordering. Section 8 proves the affirmative equiconsistency package. Section 9 and Appendix A distinguish the conclusions from the unresolved targets in the supplied report.

1 The research target and the resulting theorems

The supplied report singled out a specific gap in the cover-exacting programme: the incompatibility with a smaller extendible cardinal is known, whereas the corresponding question for a smaller strongly compact cardinal remains open in the cited notes. Its proposed next step was to locate a definability or covering condition sufficient to transfer the negative argument. We pursue exactly that step, rather than repeat the candidate survey. [1, 2]

The key distinction is between *obtaining a short cofinal set in a canonical inner model* and *making that set definable from a parameter that a relative exacting witness can absorb*. Strong compactness already supplies the first ingredient in $\text{HCD}(\delta)$. The second ingredient fails at the relevant higher completeness level. Keeping the two levels separate produces a precise conditional inconsistency theorem.

For an infinite cardinal η , $\text{CD}(\eta)$ consists of sets ordinal definable from finitely many η -complete ultrafilters on ordinals; $\text{HCD}(\eta)$ is its hereditary part. All these classes are computed in the same ambient universe. Definitions and source conventions are given in Section 2.

1.1 The definability barrier

The central statement, proved as Theorem 5.1, is

$$\text{CEx}_\gamma(\lambda) \implies \begin{cases} \text{no unbounded } a \subseteq \lambda \text{ of order type } < \lambda \text{ belongs to } \text{CD}(\gamma^+), \\ \text{cf}^{\text{HCD}(\gamma^+)}(\lambda) = \lambda. \end{cases} \quad (1)$$

The successor on γ has a concrete role: a set of size at most γ can impose at most γ many simultaneous ultrafilter constraints, and γ^+ -completeness guarantees that their intersection remains nonempty. This observation is turned into a club of small elementary substructures on which the ultrafilter becomes ordinal definable.

The statement is stronger than regularity in HOD. It rules out every short cofinal set definable from the specified complete-ultrafilter parameters, not merely every ordinal-definable one. No Axiom of Choice inside $\text{HCD}(\gamma^+)$ is assumed.

1.2 Two conditional negative conclusions

For a strongly compact $\delta < \lambda$, put

$$N_0 = \text{HCD}(\delta), \quad N_1 = \text{HCD}(\gamma^+).$$

Then $N_1 \subseteq N_0$ and Theorem 6.1 gives

$$\text{cf}^{N_0}(\lambda) < \delta < \lambda = \text{cf}^{N_1}(\lambda). \quad (2)$$

More precisely, there is an N_0 -set $a \subseteq \lambda$ of order type less than δ that has no N_1 -superset of ambient size less than λ . Consequently, either local small-subset agreement or a much weaker local covering principle between N_1 and N_0 is enough to contradict the large-cardinal configuration. These principles are stated as additional hypotheses, not inferred from strong compactness.

In the other ordering, a strongly compact $\kappa > \gamma$ provides a ground $\text{HCD}(\kappa)$ contained in $\text{HCD}(\gamma^+)$. Equation (1) therefore makes λ regular in that ground, although it has cofinality ω in V . The ground must be proper. This proves the Ground Axiom incompatibility in Corollary 7.2.

1.3 An affirmative consistency boundary

Theorem 8.3 gives an explicit theory $\mathbf{B}$ with

$$\text{Con}(\text{ZFC} + \text{I0}) \iff \text{Con}(\text{ZFC} + \mathbf{B}).$$

In a model of $\mathbf{B}$, an ultraextending λ is the first extending cardinal, $V_\lambda \subseteq \text{HOD}$, and a countable cofinal set $c \subseteq \lambda$ is outside HOD although all its intersections with HOD-sets of HOD-size less than λ are finite. Thus very extensive rank agreement is compatible with failure of precisely the covering and approximation principles that would internalize the cofinal sequence.

This comparison supplies a control on the negative arguments. Rank agreement below λ is not a substitute for covering at λ , and an inconsistency proof cannot import such covering without a separate argument.

2 Definitions, formal setting, and imported results

2.1 Ambient and internal conventions

We work in ZFC. All unadorned cardinalities, successor cardinals, clubs, and stationary sets are ambient ones. For a transitive inner model N , internal cardinality and cofinality are written $|x|^N$ and $\text{cf}^N(\alpha)$. If a is a set of ordinals belonging to two transitive models, its increasing order type $\text{ot}(a)$ is absolute between them. In particular, the assertion $\text{ot}(a) < \lambda$ does not change when we move between the models below.

For an uncountable regular η and a set A , let

$$\mathcal{P}_\eta(A) = \{\sigma \subseteq A : |\sigma| < \eta\}.$$

A club on this set is unbounded under inclusion and closed under increasing unions of nonzero length less than η . A stationary subset meets every such club. In particular, for a fixed set structure in a countable language, the elementary substructures of size less than η containing a specified finite list of elements form a club: take Skolem hulls for unboundedness and use the elementary-chain theorem for closure.

The main cover-exacting arguments involve only set-sized embeddings, so their elementarity can be expressed using satisfaction for set structures. The I0 consistency comparison is used in the standard formalization of its source. No elementarity for formulas containing an arbitrary embedding predicate is silently added.

2.2 Relative exactingness and the uniform cover bound

Definition 2.1 (Relative exacting witness). For an ordinal $\alpha > \lambda$, a λ -exacting witness relative to a set predicate $Y \subseteq V_\alpha$ is an elementary map

$$j : (X, X \cap Y) \longrightarrow (V_\alpha, Y)$$

such that $(X, X \cap Y) < (V_\alpha, Y)$, $V_\lambda \cup \{\lambda\} \subseteq X$, $\text{crit}(j) < \lambda$, and $j(\lambda) = \lambda$. Write $\text{REx}(\lambda, Y)$ when such witnesses exist at every sufficiently large height containing Y as a set.

This eventual-height convention is sufficient for every proof here. The relative property in the source notes, which requires all applicable heights, implies it. The predicate is not an arbitrary new class; it is a set and can be regarded as an element at a larger rank. The domain X need not be transitive.

Definition 2.2 (Cover exactingness). A cardinal λ is γ -cover exacting, written $\text{CEx}_\gamma(\lambda)$, if for every $\alpha > \lambda$ and $y \in V_\alpha$ there are $Y \subseteq V_\alpha$ and a relative witness as in Definition 2.1 at that height, with

$$y \in Y, \quad |Y| \leq \gamma.$$

The cardinal γ is fixed outside the height and parameter quantifiers.

We use the source's definition, including its low-height convention; only high instances are used below. We restrict attention to infinite cardinal bounds $\gamma \geq \lambda$. It is important that $y \in Y$ is membership, not inclusion, and that the witness need not fix the particular element y . [2, Definitions 2.6 and 3.1]

Fact 2.3 (Stationary relative-witness characterization). If $\text{CEx}_\gamma(\lambda)$, then, for every sufficiently large α , the set

$$\mathcal{S}_\alpha = \{\sigma \in \mathcal{P}_{\gamma^+}(V_\alpha) : \text{REx}(\lambda, \sigma)\}$$

is stationary in $\mathcal{P}_{\gamma^+}(V_\alpha)$.

Imported result. This is the implication (1) $\Rightarrow$ (3) of Blue–Goldberg’s Proposition 3.4, with the weaker eventual-height convention above. [2, Proposition 3.4] The stationary characterization is a substantive input. It is not a consequence of choosing one arbitrary cover Y .

Ordinary exactingness is obtained by omitting the predicate and cover clauses. Ultraexactingness additionally requires the witness restriction $j \upharpoonright V_\lambda$ to be an element of X . We use the equivalent cardinal-height formulations of these two properties from the supplied report and [4, 5].

2.3 Complete-ultrafilter definability

Definition 2.4. An ultrafilter is η -complete if intersections of fewer than η of its members belong to it. All ultrafilters in this paper are proper. Let $\text{UF}_\eta(\text{Ord})$ be the class of η -complete ultrafilters on ordinal underlying sets, and put

$$\text{CD}(\eta) = \text{OD}_{\text{UF}_\eta(\text{Ord})}, \quad \text{HCD}(\eta) = \text{HOD}_{\text{UF}_\eta(\text{Ord})}.$$

Here the subscript allows finitely many *members* of the indicated class as parameters, together with ordinal parameters. It does not allow arbitrary extra predicates. Let $\text{HCD} = \bigcap_\eta \text{HCD}(\eta)$.

Equivalently, a single η -complete ultrafilter parameter suffices: finite ultrafilter parameters and ordinal parameters can be combined using Fubini products and ordinal codings. We do not need this reduction, since the club argument handles finitely many parameters directly. Goldberg proves that $\text{HCD}(\eta)$ is an inner model of ZF. [3, Propositions 4.2–4.3]

Two elementary facts will be used repeatedly:

$$\eta \leq \xi \implies \text{CD}(\xi) \subseteq \text{CD}(\eta), \quad \text{HCD}(\xi) \subseteq \text{HCD}(\eta); \quad (3)$$

and, for every set of ordinals a ,

$$a \in \text{CD}(\eta) \iff a \in \text{HCD}(\eta). \quad (4)$$

For the second equivalence, once a itself is completely definable at that level, everything else in its transitive closure is an ordinal and is already ordinal definable. Monotonicity follows directly from ultrafilter completeness.

2.4 The published HCD inputs

Fact 2.5 (Goldberg). If δ is strongly compact, then $\text{HCD}(\delta)$ is a model of ZFC, is a set-forcing ground of V , and has the δ -cover property. In particular, any ambient set of ordinals of size less than δ is contained in a set of ordinals in $\text{HCD}(\delta)$ of size less than δ . If δ is extendible, then $\text{HCD}(\delta) = \text{HCD}$.

Imported result. These are Theorems 4.4, 4.10, 4.14, and 4.15 of Goldberg’s *The uniqueness of elementary embeddings*. [3] The covering conclusion just stated is weaker than the full theorem and is all that Section 6 requires. The published ground and covering theorems use *strong compactness*; the supercompact hypothesis in the shorter presentation of the lecture notes is not needed for these inputs.

A *ground* is a transitive inner model $N \models \text{ZFC}$ such that $V = N[G]$ for some set forcing $\mathbb{P} \in N$ and N -generic G . The Ground Axiom GA says that there is no proper such ground. This terminology only concerns set forcing, not arbitrary class forcing.

3 Why a short cofinal predicate is fatal

This section makes explicit the elementary obstruction used later. The local Kunen theorem is the only large-cardinal inconsistency theorem imported into these proofs. The argument is the familiar critical-sequence obstruction, with care taken over definability parameters. [6, 7, 2]

Lemma 3.1 (The critical sequence reaches the cardinal height). Let λ be a cardinal and let $e : V_\lambda \rightarrow V_\lambda$ be a nontrivial elementary embedding with critical point κ . Set

$$\kappa_0 = \kappa, \quad \kappa_{n+1} = e(\kappa_n).$$

Then $\sup_{n < \omega} \kappa_n = \lambda$. Moreover, no ordinal in $[\kappa, \lambda)$ is fixed by e .

Proof. Let $\nu = \sup_n \kappa_n \leq \lambda$. Suppose $\nu < \lambda$. Because λ is an infinite cardinal, the sequence and the ranks $V_{\nu+1}, V_{\nu+2}$ lie below the relevant height λ . The sequence belongs to V_λ , so elementarity, $e(\omega) = \omega$, and the definition of its supremum give $e(\nu) = \nu$. Restricting e therefore yields a nontrivial elementary self-embedding of $V_{\nu+2}$ with critical point below ν . The local Kunen theorem rules this out in ZFC. Hence $\nu = \lambda$.

For $\kappa \leq \xi < \lambda$, choose n with $\kappa_n \leq \xi < \kappa_{n+1}$. Then

$$e(\xi) \geq e(\kappa_n) = \kappa_{n+1} > \xi.$$

This proves the final assertion. □

In particular, such a cardinal height has ambient cofinality ω . A sufficiently high exacting witness restricts to an embedding of this kind: V_λ is definable from the fixed parameter λ , and every element of V_λ is in the source. Relativizing formulas to that set proves elementarity of the restriction. This is not an appeal to transitivity of X .

Lemma 3.2 (No fixed short cofinal set). Let $j : X \rightarrow V_\alpha$ be a sufficiently high exacting witness at λ . There is no unbounded $a \subseteq \lambda$ with $a \in X$, $j(a) = a$, and $\text{ot}(a) < \lambda$.

Proof. Let $\kappa = \text{crit}(j)$ and $\rho = \text{ot}(a)$. The order type and increasing enumeration $f : \rho \rightarrow a$ are uniquely definable from a in a sufficiently high rank. Hence $\rho, f \in X$, $j(\rho) = \rho$, and $j(f) = f$. Lemma 3.1 applied to $j \upharpoonright V_\lambda$ implies $\rho < \kappa$. Every $\xi < \rho$ is therefore fixed, and evaluation gives

$$j(f(\xi)) = j(f)(j(\xi)) = f(\xi).$$

Thus every element of a is a fixed ordinal below λ . By Lemma 3.1, all of them are below κ , contradicting unboundedness of a in λ . □

Lemma 3.3 (Relative exactingness excludes definable short cofinality). If $\text{REx}(\lambda, Y)$, no unbounded $a \subseteq \lambda$ of order type less than λ is ordinal definable from the set parameter Y .

Proof. Suppose such an a exists. Use the canonical definable well-order of OD_Y to select its least member b that is an unbounded subset of λ of order type less than λ . Crucially, b is now uniquely definable from Y and λ , without any additional free ordinal parameter.

This selection can be expressed by a single first-order formula. For example, order definition codes first by a rank in which a unique definition holds, then by the formula number and the finite ordinal-parameter tuple. The property of being the first code defining an appropriate set is a set-theoretic formula using satisfaction for *set-sized* rank structures. No truth predicate for V is being assumed.

Choose a sufficiently high V_α reflecting that formula, its uniqueness, and the relevant order-type and cofinality assertions. Also arrange that $Y, b \in V_\alpha$ and that $\text{REx}(\lambda, Y)$ supplies a witness there. Such reflecting heights exist by the usual finite-formula Reflection Theorem. In the expanded structure (V_α, Y) , the set Y is itself definable as the set of all elements satisfying the predicate. Thus $Y \in X$ and $j(Y) = Y$. Since $\lambda \in X$ is fixed, the unique definition of b implies $b \in X$ and $j(b) = b$. Lemma 3.2 gives the contradiction. $\square$

Why the ordinal parameters cause no gap

The original cofinal set may be defined using ordinals moved by a witness. We never assume those ordinals are fixed. We first select a canonical least *cofinal set*, definable from the fixed pair (Y, λ) , and only then choose a witness at a reflecting height. This is the specialized cofinality version of the definability-transfer mechanism in the cover-exacting notes.

4 Recovering an ultrafilter from a small trace

Here is the completeness calculation that ties the uniform cover bound to a specific level of the HCD hierarchy. The underlying trace idea occurs in the proof of the club-definability result in the Blue–Goldberg notes; we spell it out at the exact endpoint required here. [2, proof of Theorem 3.7]

Lemma 4.1 (Endpoint trace coding). Let η be an uncountable regular cardinal and let U be an η -complete ultrafilter on an ordinal θ . For every sufficiently large α , there is a club $C_U \subseteq \mathcal{P}_\eta(V_\alpha)$ such that

$$\sigma \in C_U \implies U \in \text{OD}_\sigma.$$

The parameter allowed on the right is the single set σ , together with ordinal parameters.

Proof. Choose α large enough to contain U , all subsets of θ , and the few power-set and coding levels used below. Let C_U be the club of $\sigma < V_\alpha$ with $U, \theta \in \sigma$ and $|\sigma| < \eta$.

Fix $\sigma \in C_U$. Since $U \cap \sigma$ has fewer than η members, η -completeness gives

$$B_\sigma = \bigcap (U \cap \sigma) \in U.$$

This set is nonempty because U is proper. Define the ordinal

$$\beta = \min B_\sigma < \theta.$$

For every $A \in \sigma$ with $A \subseteq \theta$, we have

$$A \in U \iff \beta \in A. \tag{5}$$

The forward direction is the definition of β . For the reverse direction, if $A \notin U$, then $\theta \setminus A \in U$. Elementarity of σ and $A, \theta \in \sigma$ imply $\theta \setminus A \in \sigma$, so $\beta \in \theta \setminus A$, contradicting $\beta \in A$.

We claim that U is the unique ultrafilter W on θ with $W \in \sigma$ satisfying

$$\forall A \in \sigma [A \subseteq \theta \implies (A \in W \leftrightarrow \beta \in A)]. \tag{6}$$

Existence follows from (5). For uniqueness, suppose $W \neq U$ is another such ultrafilter in σ . In V_α there is a subset $A \subseteq \theta$ on which U and W disagree. Because $U, W, \theta \in \sigma$ and $\sigma < V_\alpha$, such an A can be chosen in σ . Equation (6) would make both ultrafilters agree on A , a contradiction.

The uniqueness description uses only the set parameter σ and the ordinal parameters θ, β . It is a definition in V ; equivalently one can use satisfaction for the expanded rank structure and include α as one more ordinal parameter. Therefore $U \in \text{OD}_\sigma$. $\square$

Remark 4.2 (What is and is not being coded). The point β need not belong to σ . The ultrafilter U need not be principal. Only its *trace on subsets lying in σ* agrees with the principal ultrafilter at β . The conclusion is not that U itself is a principal ultrafilter, or that its entire membership relation is determined by one point without the parameter σ .

Corollary 4.3 (Finite parameters). If $a \in \text{CD}(\eta)$ and η is uncountable regular, then for every sufficiently large α there is a club $C \subseteq \mathcal{P}_\eta(V_\alpha)$ such that $a \in \text{OD}_\sigma$ for all $\sigma \in C$.

Proof. Choose finitely many η -complete ultrafilters $U_0, \dots, U_{m-1}$ and ordinal parameters defining a . Choose one rank high enough for all of them, and intersect the finitely many clubs supplied by Lemma 4.1. For each σ in the intersection, substitute the definitions of all U_i from σ into the definition of a . Only σ and ordinal parameters remain. $\square$

Notice the exact inequality: for $\eta = \gamma^+$,

$$|\sigma| < \eta \quad \text{means} \quad |\sigma| \leq \gamma.$$

Thus γ^+ -completeness is enough. There is no need to move to γ^{++} . Conversely, this calculation does not justify replacing γ^+ -completeness by γ -completeness when $|\sigma| = \gamma$. This is an endpoint analysis of the proof, not a claim of a model-theoretically optimal lower bound on completeness.

5 The HCD regularity barrier

Theorem 5.1 (Complete-definability barrier). Assume $\text{CEx}_\gamma(\lambda)$, where $\gamma \geq \lambda$ is a cardinal. If $a \subseteq \lambda$ is unbounded and $\text{ot}(a) < \lambda$, then

$$a \notin \text{CD}(\gamma^+).$$

Consequently λ is regular in $\text{HCD}(\gamma^+)$.

Proof. Let $\eta = \gamma^+$ and suppose $a \in \text{CD}(\eta)$ is as stated. By Corollary 4.3, at a sufficiently large rank there is a club $C \subseteq \mathcal{P}_\eta(V_\alpha)$ on which a is ordinal definable from σ . By Fact 2.3,

$$\{\sigma \in \mathcal{P}_\eta(V_\alpha) : \text{REx}(\lambda, \sigma)\}$$

is stationary. Choose σ in its intersection with C . Then $a \in \text{OD}_\sigma$ and $\text{REx}(\lambda, \sigma)$, contradicting Lemma 3.3.

Let $N = \text{HCD}(\eta)$. A cardinal of V is a cardinal in the inner model N : any inner-model bijection to a smaller ordinal would also be such a bijection in V . If λ were singular in N , it would have in N a cofinal map $f : \rho \rightarrow \lambda$ with $\rho < \lambda$. Its range a belongs to N and hence to $\text{CD}(\eta)$. Externally $|a| \leq |\rho| < \lambda$; since $a \subseteq \lambda$ and λ is a cardinal, $\text{ot}(a) < \lambda$. This contradicts the first part. Thus $\text{cf}^N(\lambda) = \lambda$. $\square$

Corollary 5.2 (Unbounded sets in the upper model are large). Under the same hypotheses, every unbounded $b \subseteq \lambda$ with $b \in \text{HCD}(\gamma^+)$ has $\text{ot}(b) = \lambda$ and $|b|^V = \lambda$. The same statement holds in place of $\text{HCD}(\gamma^+)$ for every inner model contained in it.

Proof. The order type is at most λ . Theorem 5.1 rules out every smaller order type. Its increasing enumeration therefore gives a bijection between λ and b , in the inner model and in V . For a smaller inner model, its sets are also sets of $\text{HCD}(\gamma^+)$. $\square$

5.1 An equivalent forbidden-code formulation

The first part of Theorem 5.1 can be read without any inner-model terminology. There cannot be a formula φ , a finite ordinal tuple $\vec{\xi}$, finitely many γ^+ -complete ultrafilters $\vec{U}$ on ordinals, and a unique set a such that

$$V \models \varphi(a, \vec{U}, \vec{\xi}), \quad a \subseteq \lambda, \quad \sup a = \lambda, \quad \text{ot}(a) < \lambda.$$

The prohibition concerns the *definability of the cofinal set*; the ultrafilters themselves need not be elements of the exacting witness's domain. The stationary collection of relative parameters is what bridges that gap.

5.2 A necessary failure of small covering in the ambient universe

Choose any increasing cofinal ω -sequence in λ , and let c be its range. Such a sequence exists by Lemma 3.1. If $b \in \text{HCD}(\gamma^+)$ contains c , then $b \cap \lambda$ is unbounded and belongs to the same inner model. Corollary 5.2 gives $|b|^V \geq \lambda$. Thus the upper HCD model cannot even cover this countable set by one of its sets of ambient size less than λ .

This observation alone does not refute cover exactingness. There is no theorem in use here saying that $\text{HCD}(\gamma^+)$ must have small covering in V . The role of a smaller strongly compact cardinal is to put a short cofinal set in a different, lower-completeness HCD model, making the failure a comparison between two canonical inner models.

6 A smaller strongly compact cardinal forces a cofinality gap

Theorem 6.1 (The local HCD gap). Suppose $\delta < \lambda \leq \gamma$, δ is strongly compact, and $\text{CEx}_\gamma(\lambda)$. Let

$$N_0 = \text{HCD}(\delta), \quad N_1 = \text{HCD}(\gamma^+).$$

There is an unbounded $a \subseteq \lambda$ such that

$$a \in N_0, \quad \text{ot}(a) < \delta, \quad a \notin N_1. \quad (7)$$

Moreover, for every $b \in N_1$ with $a \subseteq b$,

$$|b|^V \geq \lambda. \quad (8)$$

In particular, $\text{cf}^{N_0}(\lambda) < \delta$ but $\text{cf}^{N_1}(\lambda) = \lambda$.

Proof. Let $c \subseteq \lambda$ be the range of an increasing cofinal ω -sequence. Since δ is uncountable and N_0 has the δ -cover property by Fact 2.5, choose a set of ordinals $a_0 \in N_0$ containing c with $|a_0|^V < \delta$. Put $a = a_0 \cap \lambda$. It belongs to N_0 , is unbounded in λ , and has ambient size less than δ . As δ is a cardinal, its increasing order type is less than δ . This order type and its increasing enumeration are absolute, so the enumeration also witnesses $\text{cf}^{N_0}(\lambda) < \delta$.

By Theorem 5.1, N_1 has no unbounded subset of λ of order type less than λ . Hence $a \notin N_1$. If $b \in N_1$ contains a , then $b \cap \lambda$ is an unbounded N_1 -subset of λ . Corollary 5.2 gives $|b \cap \lambda|^V = \lambda$, which implies (8). The remaining cofinality assertion is Theorem 5.1. $\square$

6.1 A local covering principle that is already too much

Definition 6.2 (Generous local covering). For inner models $N_1 \subseteq N_0$ and cardinals $\delta < \lambda$, let $\text{Cov}_\lambda^{\delta, < \lambda}(N_1, N_0)$ denote the assertion

$$\forall a \in N_0 \cap [\lambda]^{< \delta} \exists b \in N_1 \quad a \subseteq b \subseteq \lambda \wedge |b|^V < \lambda. \quad (9)$$

Both sizes in this formula are measured in the ambient universe.

This is much less demanding than agreement on all subsets of λ . Even the permitted cover size may grow from less than δ to any cardinal less than λ . The result does not depend on insisting on exact cardinal preservation between the two inner models.

Corollary 6.3 (Conditional inconsistency above a strongly compact cardinal). The following theory is inconsistent:

$$\text{ZFC} + \exists \delta < \lambda \leq \gamma \left[\text{SC}(\delta) \wedge \text{CE}_{\gamma}(\lambda) \wedge \text{Cov}_\lambda^{\delta, < \lambda}(\text{HCD}(\gamma^+), \text{HCD}(\delta)) \right]. \quad (10)$$

The same conclusion holds with the covering clause replaced by

$$\text{HCD}(\delta) \cap [\lambda]^{< \delta} \subseteq \text{HCD}(\gamma^+). \quad (11)$$

Proof. Apply (9) to the set a from Theorem 6.1. The resulting b contradicts (8). Condition (11) implies (9) by choosing $b = a$. $\square$

Ordinary δ -covering of N_0 by N_1 , with covers of N_1 -size less than δ , also implies (9) for the relevant small sets whenever it applies to them. More precisely, it suffices that it apply to sets that have N_0 -size less than δ ; the particular a in the proof has such size because its order type is less than δ in N_0 . Thus the theorem rules out the standard internal version as well, without assuming that all ambient-small N_0 -sets are internally small.

6.2 Recovering the extendible prohibition

Corollary 6.4 (Known boundary recovered). An extendible δ cannot lie below a γ -cover-exacting λ .

Proof. An extendible cardinal is strongly compact. By Fact 2.5, $\text{HCD}(\delta) = \text{HCD}$. Since $\delta < \gamma^+$, monotonicity gives

$$\text{HCD}(\delta) = \text{HCD} \subseteq \text{HCD}(\gamma^+) \subseteq \text{HCD}(\delta).$$

Hence the two models are equal and (11) holds, contrary to Corollary 6.3. $\square$

This is a recovery of Blue–Goldberg’s Theorem 3.6, not a second claim to that result. [2] The new formulation separates what is used from strong compactness—a short cofinal set in $\text{HCD}(\delta)$ —from what extendibility adds—stabilization of the complete-definability hierarchy.

6.3 Exactly what is missing in the unrestricted problem

Strong compactness by itself supplies δ -covering from V into $\text{HCD}(\delta)$. It does *not*, by the theorem cited here, supply covering from $\text{HCD}(\delta)$ into $\text{HCD}(\gamma^+)$. These are different pairs of models. Using the former theorem as though it were the latter would erase the essential gap in (2).

A resolution by this route would require proving a principle such as (9) from the original coexistence hypotheses, or obtaining one forbidden short cofinal code in $\text{CD}(\gamma^+)$ by another method. The present proof does neither. Its output is the explicit incompatibility (10) and the necessary, sharply witnessed failure of covering in every putative model of the unrestricted coexistence theory.

7 A larger strongly compact cardinal forces a proper ground

The same regularity theorem gives an incompatibility with a familiar axiom that does not mention HCD. Here the strongly compact cardinal lies above the *cover bound*, not below λ .

Theorem 7.1 (Ground singularization). Suppose $\lambda \leq \gamma < \kappa$, $\text{CEx}_\gamma(\lambda)$, and κ is strongly compact. Then

$$N = \text{HCD}(\kappa)$$

is a proper set-forcing ground of V , and

$$\text{cf}^N(\lambda) = \lambda, \quad \text{cf}^V(\lambda) = \omega. \quad (12)$$

For every presentation $V = N[G]$ with forcing $\mathbb{P} \in N$,

$$|\mathbb{P}|^N \geq \lambda. \quad (13)$$

Proof. Because κ is a cardinal greater than γ , it is at least γ^+ . By monotonicity,

$$N = \text{HCD}(\kappa) \subseteq \text{HCD}(\gamma^+).$$

Fact 2.5 says that $N \models \text{ZFC}$ and that N is a set-forcing ground of V . A cofinal map in N from an ordinal less than λ into λ would belong to $\text{HCD}(\gamma^+)$ and contradict Theorem 5.1. Thus λ is regular in N . Lemma 3.1 gives its ambient cofinality ω , so $N \neq V$.

For the size assertion, suppose $\mu = |\mathbb{P}|^N < \lambda$. In N , choose a name $\dot{f}$ and a condition $p \in G$ forcing that $\dot{f} : \omega \rightarrow \lambda$ is cofinal. For each $n < \omega$, choose a maximal antichain below p deciding $\dot{f}(n)$. Each antichain has at most μ elements, and hence there are at most μ possible values for each n . Let $B \in N$ be the union of these sets of possible values. Inside N ,

$$|B| \leq \mu \cdot \aleph_0 < \lambda.$$

Regularity of λ in N implies $\sup B < \lambda$. But p forces $\text{ran}(\dot{f}) \subseteq B$, contradicting cofinality. This proves (13). $\square$

Corollary 7.2 (Ground Axiom incompatibility). The theory

$$\text{ZFC} + \text{GA} + \exists \lambda \leq \gamma < \kappa [\text{CEx}_\gamma(\lambda) \wedge \text{SC}(\kappa)] \quad (14)$$

is inconsistent. In particular, the Ground Axiom together with a proper class of strongly compact cardinals excludes every cover-exacting cardinal with a set-sized uniform cover bound.

Proof. Theorem 7.1 supplies a proper ground, contradicting GA. With a proper class of strongly compact cardinals, choose one above any given bound γ . $\square$

This is a conditional incompatibility among explicitly named principles. It does not imply that cover exactingness and a larger strongly compact cardinal are inconsistent: it implies that their coexistence requires a nontrivial ground. The ground need not be HOD, and the forcing witnessing it need not be Prikry forcing. The proof only determines its effect on $\text{cf}(\lambda)$ and the lower bound (13).

There is also a useful consistency check on the ordering. The cover-exacting notes describe constructions with cover-exacting cardinals below larger large cardinals. [2, discussion following Theorem 3.5] Nothing here contradicts those constructions; a nontrivial forcing extension is precisely the sort of universe the theorem permits.

8 An I0-equiconsistent sharp HOD boundary

The preceding theorems are negative only when a specified covering or geology assumption is added. We now make the positive side equally explicit. This section uses an existing forcing construction and derives a stronger package of boundary properties from its output. It does not claim a new construction of an ultraexacting cardinal.

8.1 The two consistency inputs

Aguilera–Bagaria–Goldberg–Lücke prove both

$$\text{Con}(\text{ZFC} + \exists \lambda \text{ UEx}(\lambda)) \longleftrightarrow \text{Con}(\text{ZFC} + \text{I0}) \quad (15)$$

and the following preservation-and-coding result: from an ultraexacting λ , forcing can preserve its ultraexactingness and V_λ while making $V_\lambda \subseteq \text{HOD}$. [5, Theorem 4.5; Proposition 3.9] We also use the established theorem that an exacting λ is regular in HOD. [4, Theorem 2.10]

The consistency transformations in (15) need not preserve the ordinal at which the principle holds. Our equiconsistency conclusion below does not assert otherwise.

8.2 The boundary theory

Definition 8.1. Let $\mathbf{B}$ assert that there are a cardinal λ and an increasing cofinal ω -sequence with range $c \subseteq \lambda$ satisfying all of the following:

1. λ is ultraexacting and there is no exacting cardinal below λ .
2. $V_\lambda \subseteq \text{HOD}$, but $c \notin \text{HOD}$. In particular, $V_{\lambda+1} \not\subseteq \text{HOD}$.
3. λ is regular in HOD, and for every $a \in \text{HOD}$ with $|a|^{\text{HOD}} < \lambda$, the intersection $c \cap a$ is finite and belongs to HOD.
4. Every $b \in \text{HOD}$ with $c \subseteq b$ has $|b|^V \geq \lambda$.

All cardinalities explicitly superscripted by HOD are computed internally. The final cardinality is an ambient one. These two conventions are not interchangeable in a general inner-model comparison.

Lemma 8.2 (A single cofinal set supplies all the boundary witnesses). Suppose λ is exacting and $V_\lambda \subseteq \text{HOD}$. Then any range c of an increasing cofinal ω -sequence in λ satisfies items (2)–(4) of Definition 8.1, and there is no exacting cardinal below λ .

Proof. Let $H = \text{HOD}$. The cited exacting regularity theorem gives $\text{cf}^H(\lambda) = \lambda$, whereas Lemma 3.1 gives $\text{cf}^V(\lambda) = \omega$. If $c \in H$, its increasing enumeration would be a cofinal map $\omega \rightarrow \lambda$ in H , a contradiction. Since c is unbounded in λ , its rank is exactly λ . Thus it belongs to $V_{\lambda+1} \setminus H$, while every set of smaller rank is in H by assumption. This proves item (2) and the regularity part of (3).

Now let $a \in H$ with $|a|^H < \lambda$. The set $a \cap \lambda$ is in H and has H -size less than λ . Regularity in H makes it bounded in λ . An increasing ω -sequence cofinal in λ has only finitely many terms below any particular bound. Hence $c \cap a$ is finite. Every finite set of ordinals is in HOD , using its finitely many members as ordinal parameters. This proves (3).

If $b \in H$ contains c , the set $d = b \cap \lambda$ is an unbounded H -subset of λ . Its order type cannot be less than λ , by regularity in H , and cannot exceed λ , since $d \subseteq \lambda$. Thus $\text{ot}(d) = \lambda$. The increasing enumeration is an actual bijection between λ and d , so $|b|^V \geq \lambda$, proving (4).

Finally, suppose $\mu < \lambda$ were exacting. It would have ambient cofinality ω and be regular in HOD . The range of an increasing cofinal ω -sequence in μ has rank $\mu < \lambda$, so it belongs to $V_\lambda \subseteq H$. Its increasing enumeration in H contradicts the regularity of μ there. Therefore no such μ exists. $\square$

Theorem 8.3 (Equiconsistency of the full boundary package).

$$\text{Con}(\text{ZFC} + \mathbf{B}) \iff \text{Con}(\text{ZFC} + \mathbf{I0}).$$

Proof. For the forward implication, $\mathbf{B}$ includes the existence of an ultraexacting cardinal. Apply the corresponding direction of (15).

Conversely, the other direction of (15) provides the relative consistency of a model of ZFC with an ultraexacting λ . Apply the cited preservation-and-coding forcing to obtain a model in which λ remains ultraexacting and $V_\lambda \subseteq \text{HOD}$. Choose an increasing cofinal ω -sequence in λ in that model. Lemma 8.2 verifies every additional clause of $\mathbf{B}$. Thus the original consistency assumption gives $\text{Con}(\text{ZFC} + \mathbf{B})$. $\square$

The first-exacting conclusion and the optimal rank agreement are already present in the cited paper's discussion and Corollary 3.10. [5] The role of the formulation above is to include a single, explicit witness to all the approximation and covering failures, and to record the consistency strength of the whole package. It should not be mistaken for an independent proof of (15).

8.3 Approximation failure, with the quantifiers visible

For inner models $N \subseteq W$, the ρ -approximation property says that whenever $A \in W$, $A \subseteq N$, and

$$A \cap a \in N \quad \text{for every } a \in N \text{ with } |a|^N < \rho,$$

then $A \in N$. It is customary to study this for regular uncountable ρ , but the displayed condition makes sense at any cardinal parameter.

In a model of $\mathbf{B}$, the same set $c \subseteq \lambda \subseteq \text{HOD}$ satisfies

$$c \notin \text{HOD}, \quad \forall a \in \text{HOD} [|a|^{\text{HOD}} < \lambda \implies c \cap a \in \text{HOD}]. \quad (16)$$

It follows that (HOD, V) fails the ρ -approximation property for every ambient regular uncountable $\rho < \lambda$. Equation (16) also witnesses failure of the displayed approximation condition at λ itself, without calling the ambiently singular λ regular.

Likewise, item (4) prevents HOD from covering the countable set c by any of its sets of ambient size less than λ . In particular, the usual ρ -cover property fails for every ambient regular uncountable $\rho < \lambda$. An internally small cover would also be externally small, so the internal-size convention cannot avoid this conclusion.

8.4 What the positive package does not imply

It does not put a short cofinal map into HOD. It does not imply agreement on subsets of V_λ . It does not make an arbitrary ambient well-order of $V_{\lambda+1}$ ordinal definable. And it does not turn an ultraexacting cardinal into a cover-exacting cardinal with any particular cover bound. The affirmative construction therefore neither contradicts Theorem 5.1 nor solves the smaller-strongly-compact coexistence question.

Its exact boundary is useful: every set of rank *less than* λ may be in HOD, but one cannot extend that inclusion to all sets of rank λ . The countable cofinal range already supplies a counterexample.

9 What has been established, and what remains open

9.1 The contribution relative to the supplied report

The supplied report proposed locating a local definability condition behind the extendible obstruction. The present work gives such a condition with explicit cardinal parameters. It proves regularity at the particular HCD level γ^+ , extracts a short cofinal set from the smaller strongly compact cardinal's HCD model, and rules out even generous local covering between those two levels. The opposite ordering produces a further named-axiom incompatibility with the Ground Axiom.

The core statements are

$$\begin{aligned} \text{CEx}_\gamma(\lambda) &\Rightarrow \text{cf}^{\text{HCD}(\gamma^+)}(\lambda) = \lambda, \\ \text{SC}(\delta) \wedge \delta < \lambda \wedge \text{CEx}_\gamma(\lambda) &\Rightarrow \neg \text{Cov}_\lambda^{\delta, < \lambda}(\text{HCD}(\gamma^+), \text{HCD}(\delta)), \\ \text{SC}(\kappa) \wedge \gamma < \kappa \wedge \text{CEx}_\gamma(\lambda) &\Rightarrow \neg \text{GA}. \end{aligned}$$

The proofs are given above, with imported results isolated in Section 2. These are mathematical consequences, not proposals for future lemmas. Their dependence on the lecture-note stationary characterization is explicit.

9.2 Novelty and verification status

The bounded HCD regularity, local covering obstruction, and ground singularization statements are derived in this follow-up from established techniques. I did not locate these exact formulations in the primary texts checked. That is not a certification that no equivalent result is known; some or all may already be recognized corollaries by specialists. The ultrafilter trace argument is credited to the notes, and the complete definability, covering, and ground theorems are credited to Goldberg.

The I0 theorem here is deliberately described as an equiconsistency *refinement/package*: the nontrivial consistency construction is already in Aguilera–Bagaria–Goldberg–Lücke. No historical novelty is claimed for the forcing construction, the first-exacting corollary, or the impossibility of extending $V_\lambda \subseteq \text{HOD}$ to $V_{\lambda+1}$.

The arguments have been checked at the level of ordinary mathematical proofs, including the completeness endpoint, ordinal parameters, internal versus external size, and the direction of covering. They have not been independently refereed or formalized in a proof assistant. Typesetting compilation is not presented as mathematical machine verification.

9.3 The remaining mathematical obstruction

The theory

$$\text{ZFC} + \exists \delta < \lambda \leq \gamma [\text{SC}(\delta) \wedge \text{CE}_{\gamma}(\lambda)]$$

is not refuted by this paper. Under that theory, the HCD tower must have the specific cofinality gap in Theorem 6.1. To obtain an unconditional refutation by this route, one must show that this gap is impossible using only the original hypotheses. The published covering theorem does not do that: it covers into the lower-completeness model, whereas the contradiction needs a cover in the higher-completeness model.

Conversely, a positive model of the theory would necessarily contain the short set a in (7) and exhibit its failure to acquire any small cover at the upper HCD level. This is a concrete requirement on a construction, not just the vague requirement that HOD be far from V .

The global cardinal-preserving embedding route was also checked against the primary literature. Reconstructing the embedding range from the external stationarity pattern of a transported partition is already part of Hamkins–Kirmayer–Perlmutter’s argument. [7, proofs of Theorems 11–12] It is therefore not offered here as a new result. No new inconsistency of $\text{I0}^\#$, ordinary exacting–extendible coexistence, or HOD–Berkeley cardinals is asserted.

Bottom line

There is a proved local inconsistency theorem, a proved Ground Axiom incompatibility, and a fully specified equiconsistency corollary. The main newly articulated boundary is the jump from short cofinality in $\text{HCD}(\delta)$ to regularity in $\text{HCD}(\gamma^+)$, with an actual set witnessing the failure of every cover of size less than λ . Removing that covering failure from a putative coexistence model would require additional mathematics; it is not supplied by changing the name of the compactness axiom.

A Dependency and hypothesis audit

Conclusion	Exact additional input	Potential invalid substitution avoided
Critical sequence cofinal in λ	Local Kunen theorem in ambient ZFC; λ a cardinal.	A rank- $\lambda + 1$ embedding is not refuted. The forbidden restriction occurs only if the critical limit lies strictly below the cardinal height.
Relative definable cofinality contradiction	Witnesses at arbitrarily high reflecting ranks; canonical least cofinal set.	Arbitrary ordinal parameters of an original definition are not presumed fixed.
Trace coding	η regular; $ \sigma < \eta$; U η -complete; $\sigma < V_\alpha$.	Principal behavior on a small trace is not principality of U . The coding point need not lie in σ .
Regularity in $\text{HCD}(\gamma^+)$	Stationary relative-witness characterization, not merely one cover.	Internal stationarity is not used as ambient stationarity. No Choice in this HCD level is assumed.
Short cofinal set in $\text{HCD}(\delta)$	$\delta < \lambda$ strongly compact; the published δ -cover theorem.	Covering into $\text{HCD}(\delta)$ is not covering into $\text{HCD}(\gamma^+)$.
Local inconsistency	Local covering or local small-set agreement between the two specified HCD levels.	That extra principle is not inferred from strong compactness.
Extendible contradiction	Published $\text{HCD}(\delta) = \text{HCD}$ for extendible δ .	This recovers a known theorem; it is not a new refutation of ordinary exactingness.
Proper-ground theorem	A strongly compact $\kappa > \gamma$, with γ the fixed cover bound.	The order $\delta < \lambda$ from the other theorem is not reused. The ground is HCD, not necessarily HOD.
Equiconsistency package	Published ultraextending/I0 comparison and high coding that preserves ultraextendingness.	The same ordinal need not witness I0 in the consistency transformation. No new forcing proof is claimed.
Approximation witness	HOD regularity at λ , a strictly increasing cofinal ω -sequence, and internal smallness of test sets.	Ambiently singular λ is not called regular; the witness formula is stated directly.

A.1 The core dependency chain

The proof of the principal negative result can be checked in the following order:

$$\begin{aligned}
&\text{local Kunen} \implies \text{Lemma 3.1} \implies \text{Lemmas 3.2–3.3}, \\
&\text{ultrafilter completeness} \implies \text{Lemma 4.1} \implies \text{Corollary 4.3}, \\
&\text{Fact 2.3} + \text{Lemma 3.3} + \text{Corollary 4.3} \implies \text{Theorem 5.1}, \\
&\text{Theorem 5.1} + \text{strongly compact HCD covering} \implies \text{Theorem 6.1}, \\
&\text{Theorem 6.1} + \text{local covering} \implies \perp.
\end{aligned}$$

The geology theorem replaces the last two lines by the published HCD ground theorem at a strongly compact cardinal above γ . The affirmative I0 package is independent of this dependency chain.

B Source and reproducibility notes

The starting document was the supplied `Large_Cardinals_Unified_Report.tex`. Its preamble was reused for the body and mathematical fonts, page geometry, heading hierarchy, color values, running furniture, and callout style. The new document does not reproduce the earlier candidate survey.

The primary sources were checked on 18 September 2026. In particular, Goldberg’s published HCD theorem was checked directly rather than relying on the stronger hypothesis in its short lecture-note presentation. The cover-exacting stationary characterization remains attributed to the July 2026 lecture notes. The ABGL results are cited to arXiv version 1; the ABL regularity theorem is cited to version 4.

The archive contains this source, its compiled PDF, and a README with build instructions and the verification scope. The source is self-contained apart from standard LaTeX packages. Compile twice with pdfLaTeX, or use `latexmk -pdf`. The required body/math packages are `newpxtext` and `newpxmath`; the sans-serif accents follow the supplied preamble. No external figures, bibliography database, or custom font files are required.

References

- [1] *Large cardinals at the inconsistency frontier: A unified assessment of ZFC-compatible hypotheses, their obstructions, and the evidence on both sides*. Report supplied by the user, dated 18 September 2026. Source: `Large_Cardinals_Unified_Report.tex`. Sections 6 and 13 give the cover-exacting target and proposed theorem-transfer programme.
- [2] G. Goldberg. *Consistency beyond the Kunen inconsistency*. Lecture notes dated 1 July 2026, reporting joint work with D. Blue. <https://math.berkeley.edu/~goldberg/Slides/CoverExact.pdf>. Definitions 2.6, 3.1, and 3.2; Lemma 3.3; Proposition 3.4; Theorems 3.5–3.8; Problem 5.2. The stationary characterization and trace-coding mechanism are the inputs used here. Cited as lecture notes, not as a refereed cover-exacting paper.
- [3] G. Goldberg. *The uniqueness of elementary embeddings*. Journal of Symbolic Logic **89**(4) (2024), 1430–1454. Author manuscript: <https://math.berkeley.edu/~goldberg/Papers/UniquenessOfElementaryEmbedding.s.pdf>. Propositions 4.2–4.3; Theorems 4.4, 4.10, 4.14, and 4.15. The ground and covering assertions require strong compactness; the stabilization assertion uses extendibility.
- [4] J. P. Aguilera, J. Bagaria, and P. Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. [arXiv:2411.11568v4](https://arxiv.org/abs/2411.11568), 16 September 2025. Theorem 2.10 supplies regularity in HOD_{V_λ} , and hence in HOD, for an exacting cardinal λ .
- [5] J. P. Aguilera, J. Bagaria, G. Goldberg, and P. Lücke. *Large cardinals beyond HOD*. [arXiv:2509.10254v1](https://arxiv.org/abs/2509.10254), 12 September 2025. Proposition 3.9, Corollary 3.10, and Theorem 4.5 supply the high coding, first-exacting control, and ultraexacting/I0 equiconsistency used in Section 8.
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